MAHMOUD ABD EL-GHAFFAR

AI Automation Engineer & Systems Integration Specialist

Make | Zapier | n8n | Custom REST APIs | Webhooks | Multi-Agent Workflows

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PROFESSIONAL SUMMARY

Accomplished Computer Engineer and AI Automation Specialist with deep hands-on expertise engineering production workflows, multi-agent pipelines, and cloud integrations across Make, Zapier, n8n, and custom REST APIs. Built and co-built 30+ native integration nodes inside the AI MicroMind core runtime and automated 30+ end-to-end business workflows connecting CRMs (HubSpot, Airtable), web forms, messaging platforms (Slack, Gmail, WhatsApp/WAPA), and enterprise databases. Known for strong systems thinking, proactive error handling, and the ability to articulate technical architectures in plain, non-technical language for business stakeholders. Comfortable working under strict NDA and proprietary client data privacy standards.

CORE TECHNICAL COMPETENCIES

- **Automation Platforms & Orchestration:** Make (Integromat), Zapier, n8n, AI MicroMind (30+ Native Core Nodes), Webhooks, Cron Schedulers, REST APIs, OAuth2.
- **Business & CRM Integrations:** HubSpot, Airtable, Gmail, Microsoft 365, Slack, Meta/WhatsApp (WAPA), Telegram, ClickUp, Teamwork, ConnectWise, Typeform.
- **AI & Multi-Agent Frameworks:** LangChain (Expert), LangGraph (Advanced), Model Context Protocol (MCP), Pinecone Vector DB, OpenAI Embeddings, LLM-to-SQL.
- **Systems Reliability & Delivery:** Systems Architecture, Automated Error Handling & Retries, Real-Time Alerting, SOPs & Plain-Language Client Documentation.
- **Enterprise ERPs & Database BI:** Conversational BI over Oracle ERP, Oracle Fusion, Dynamics 365 OData, SAP, Odoo, PostgreSQL, SQL Server.
- **Backend, Cloud & Acceleration:** Node.js / TypeScript, Python, Docker, AWS, DigitalOcean Linux VPS, Render, GitHub CI/CD, Google Antigravity, Claude Code.

PROFESSIONAL EXPERIENCE

AI & Automation Engineer — AI MicroMind

2024 – Present

- **Native Integration Nodes:** Built and co-built 30+ native integration nodes directly inside AI MicroMind Core (Node.js runtime) for WAPA (WhatsApp API), Telegram, Gmail, Weather, MCP Client, Pinecone, Webhooks, Schedulers, and PostgreSQL Audit Loggers using AI coding tools (Antigravity, Claude Code, Codex).
- **Multi-Step Business Automations:** Designed automated workflows across Zapier, Make, and n8n connecting CRMs (HubSpot, Airtable), web forms, and messaging tools, eliminating manual data entry and accelerating client operational pipelines.
- **Conversational BI & Report Bot:** Architected the AI MicroMind App featuring the Report Bot Framework, enabling non-technical users to query on-premise ERPs (Oracle, SAP, Microsoft Dynamics, Odoo, SQL) and cloud databases via natural language.
- **Proactive Error Handling & 99.9% Uptime:** Implemented robust fallback triggers, alert notifications, and webhook retry mechanisms, ensuring zero data loss and fault-tolerant production execution.
- **Client Communication & Plain-Language SOPs:** Authored client-facing visual workflow diagrams, SOPs, and user guides, translating intricate technical pipelines into plain-language instructions for non-technical stakeholders.
- **Multi-Agent Graphs & MCP:** Constructed multi-agent graph workflows using LangChain and LangGraph with Model Context Protocol (MCP) server dispatchers for secure backend script execution.

PROFESSIONAL EXPERIENCE (CONTINUED)

Automation Engineer — Operatives (IT Services)

Jun 2021 – Jan 2022

- **Workflow Synchronization:** Analyzed cross-departmental business workflows and engineered automated data synchronization pipelines, eliminating repetitive manual bottlenecks and reducing data-entry errors by over 70%.
- **Integration Reliability:** Maintained multi-system integrations across internal service catalogs, ticketing systems, and database stores with automated error logging and alerting.

Operation Engineer & Admin — Operatives (IT Services)

Jan 2020 – Jun 2021

- **IT Infrastructure & Operations:** Administered IT infrastructure operations, monitored workflow execution logs, enforced strict SLA compliance, and resolved technical service tickets with prompt escalation and root-cause analysis.
- **System Health & Maintenance:** Conducted regular preventative maintenance, scheduled automated backup routines, and drafted operational runbooks for technical staff.

KEY TECHNICAL ARCHITECTURE & INTEGRATION PROJECTS

- **AI MicroMind App & Report Bot Framework:** Conversational BI reporting engine translating natural language prompts into live SQL/OData queries across Oracle, SAP, Dynamics 365, Odoo, and SQL databases.
- **AI MicroMind Core Node Architecture (30+ Native Nodes):** Custom Node.js native nodes extending core platform capabilities for WAPA, Telegram, Gmail, Weather, MCP, Pinecone, and Cron Schedulers.
- **End-to-End Business Automation Pipelines:** Multi-platform automations across Make, Zapier, and n8n connecting web forms, HubSpot/Airtable CRMs, email sequences, and Slack notifications with automated error alerts.
- **AI MicroMind MicroApps & Mobile Architecture:** Modular middleware connecting cross-platform mobile apps and 12+ SaaS platforms (Lark, Bubble.io, Teamwork, ConnectWise, ClickUp, Zoom, Slack).
- **Stateful Multi-Agent Graphs (LangChain & LangGraph):** Hierarchical supervisor agent graphs featuring state persistence, conditional branching, and self-correcting prompt loops.
- **Cloud DevOps & Containerization Suite:** Multi-stage Dockerized environments deployed across AWS, DigitalOcean Linux VPS, and Render with automated CI/CD.

EDUCATION

B.Sc. in Computer Engineering — Faculty of Engineering Shoubra, Benha University

Benha, Egypt

- **Specialization:** Computer Systems Architecture, Database Management Systems, Software Engineering & AI Systems.

PROFESSIONAL STANDARDS & WORKING STYLE

- **Plain-Language Client Communication:** Skilled in communicating complex technical workflows and automation logic in simple, plain language to non-technical business clients and non-technical stakeholders.
- **Data Privacy & Strict NDA Adherence:** Experienced and completely comfortable working under strict Non-Disclosure Agreements (NDA) and safeguarding sensitive proprietary client data.
- **Autonomous Troubleshooting & Proactivity:** Proactive engineer skilled in identifying edge cases, tracing webhook logs, and troubleshooting integration breaks independently without hand-holding.